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Applied Scientist / Scientific ML & HPC (Computational Chemistry / DFT)

Large-scale training pipelines, differentiable modeling, GPU performance under memory constraints.

Summary

Applied scientist with 10+ years in computational chemistry (DFT) and scientific machine learning. I build production-grade training and benchmarking pipelines for physics-based models, with a strong focus on performance, numerical stability, and reproducibility under compute and memory constraints. Experienced in custom TensorFlow models (non-standard layers and training loops), multi-GPU execution on HPC (SLURM), and integration with externally maintained production simulation codes. Publication track record: 12 papers (published / in press) over the last 2 years.

Selected Highlights

- **Scale:** 10k–30k reactions per training cycle; 1,000–1,500 batches; 3–6M grid points per batch.
- **Throughput engineering:** developed a streaming data loader with CPU prefetch, overlapping I/O and GPU compute.
- **Multi-GPU:** 1.7–1.8× practical speedup on 2 GPUs in production-like runs.
- **Accuracy (DFT):** LH25nP-D4 achieved ~30% lower benchmark error (WTMAD-2) (2.5 vs. 3.5 kcal/mol) than the widely used ω B97M-V on the GMTKN55 suite (published; implemented in Turbomole).

Core Competencies

- **Scientific ML:** TensorFlow, custom layers, custom training/eval loops, large-batch optimization.
- **HPC / Performance:** multi-GPU training, memory-aware batching, profiling-driven optimization, SLURM stability.
- **Data & Pipelines:** dataset generation, streaming loaders, batching, aggregation from grid points to molecule/reaction targets.
- **Computational Chemistry (DFT):** method development and benchmarking; Turbomole integration.

Professional Experience

Researcher / Applied Scientist (Academic R&D)

2016–Present

Technische Universität Berlin, Germany

- **Milestone-driven end-to-end ownership (DFT + Scientific ML):** defined roadmap and acceptance criteria (accuracy vs. cost) and delivered 4 milestone releases in the last 18 months, including public releases implemented in Turbomole.
- **Method development:** developed ML-enhanced DFT models (local hybrids and extensions), including physics-informed choices for difficult regimes (e.g., delocalization vs. strong-correlation trade-offs).
- **Training & benchmarking workflows:** built repeatable grid-based training/evaluation pipelines and benchmark checks for consistent comparisons across releases.

- **Data & performance engineering:** implemented a molecule-level on-disk cache for reusable artifacts and tuned batch sizing/update frequency under VRAM constraints to maximize throughput on SLURM.
- **Integration & knowledge transfer:** integrated with Turbomole, maintained interoperability across toolchains, and mentored a PhD student for ownership transfer.

Research Assistant

2013–2014

Institute of Nuclear Chemistry and Technology, Poland
Computational studies of actinide/lanthanide selectivity.

PhD Researcher

2011–2015

University of Warsaw, Poland
Computational protocols for transition-metal NMR; teaching and mentoring.

Selected Technical Tools

- **Programming:** Python, Fortran, Bash, awk
- **ML:** TensorFlow (custom models, training loops, evaluation/benchmarking)
- **HPC:** SLURM, OpenMP, MPI, Linux; profiling and stability practices for large runs
- **Scientific software:** Turbomole, PySCF, Gaussian, DIRAC, ADF

Education

PhD in Chemical Science — University of Warsaw, Poland
MSc in Chemistry — University of Warsaw, Poland

2011–2015

2006–2011

Grants

Mobility Plus Fellowship — Polish Ministry of Science
 Postdoctoral project on iron-sulfur clusters in proteins (TU Berlin).

2016–2017

Languages

Polish (Native), English (C1), German (A2)

Code Availability

Selected repositories and examples are available via the portfolio/GitHub link above; additional code (including GPU-optimized training components) is available upon request.